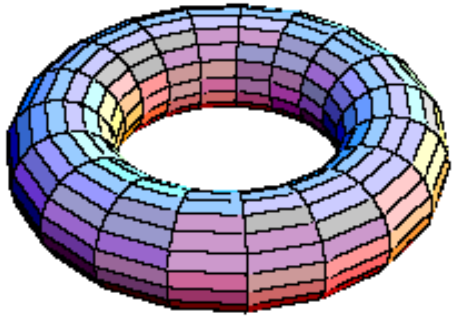
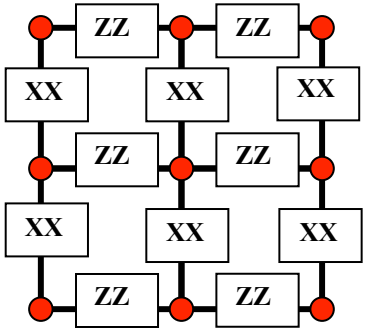
I EEE A O OOA

Topological Codes and Subsystem Codes and Why We Should Care About Them…

Dave Bacon
Department of Computer Science & Engineering
Unive ttle, WA











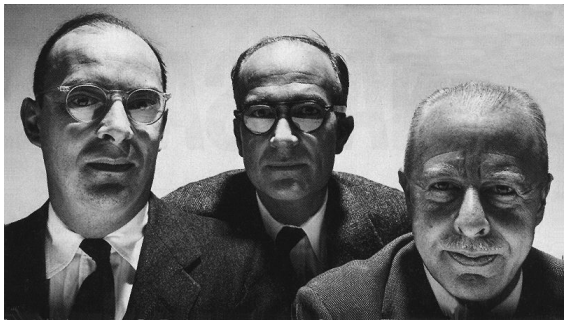




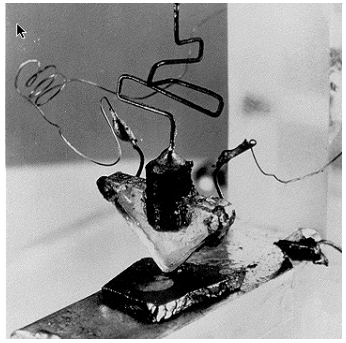




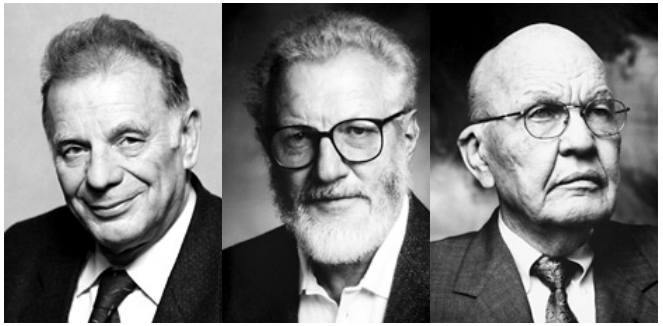
Lessons



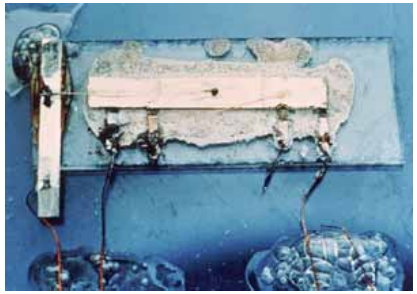
1956
Shockley, Bardeen, Brattain



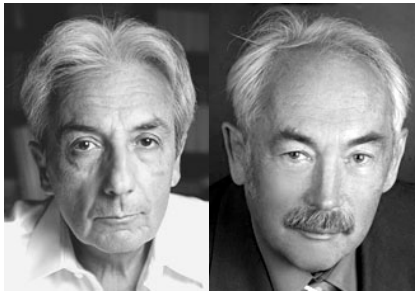
Transistor



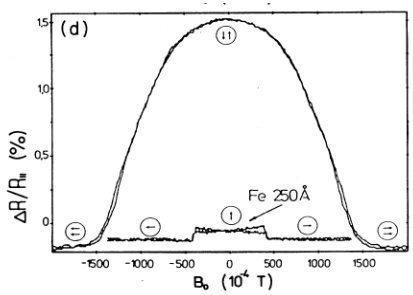
2000
Alferov, Kroemer, Kilby



Integrated Circuit



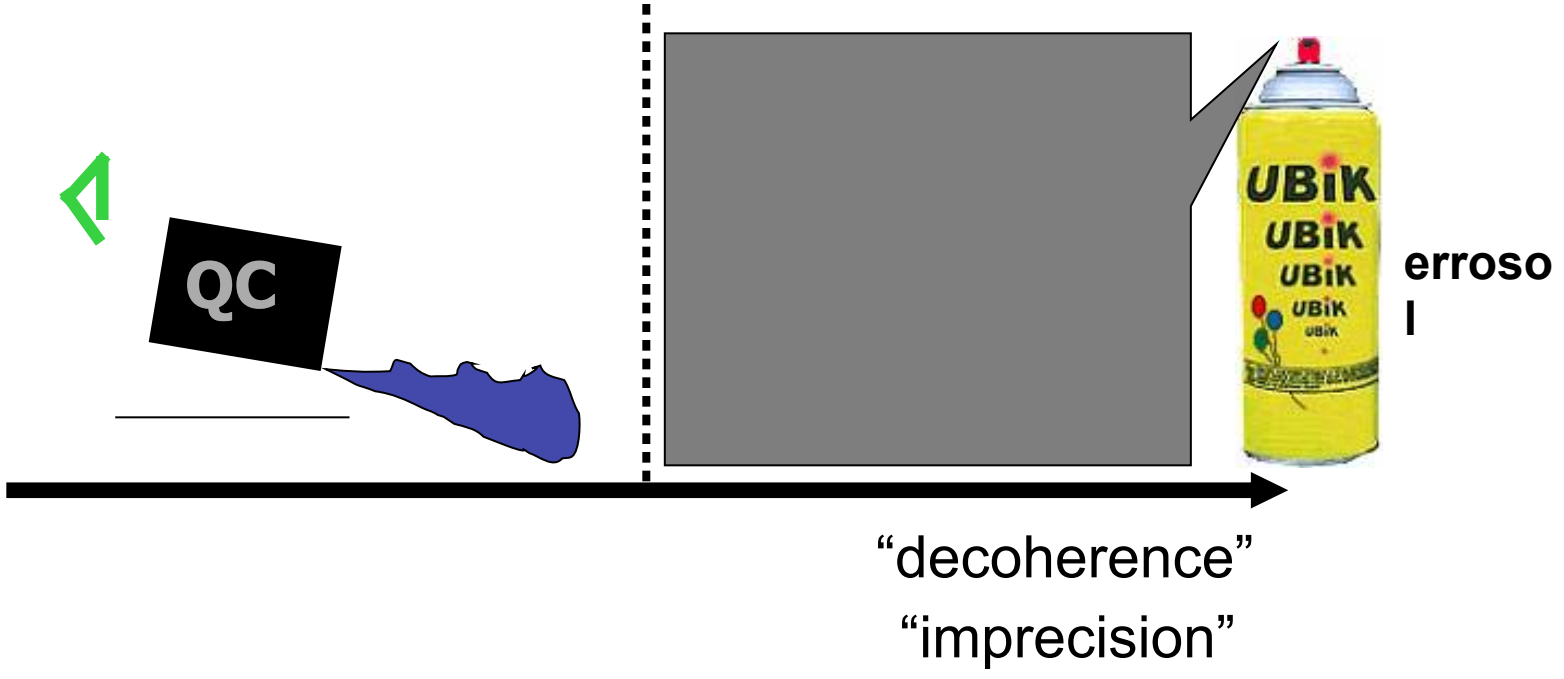
2007
Fert, Grünberg



Giant Magnetoresistance

Can We Make Physical Devices Which Robustly Quantum Compute?

Our Panacea: Threshold Theorem



Threshold Theorem for Fault-Tolerant Quantum Computation

[Aharonov, Ben-Or, Knill, LaFlamme, Zurek, Kitaev]

If noise is weak enough and control is precise enough and of the “correct” form, then a quantum computation to with a desired error can be carried out with an overhead polynomial in the log of one over the error.

Can We Make Physical Devices Which Robustly Quantum Compute?

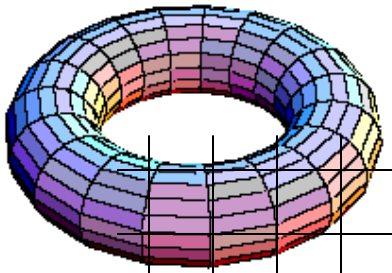


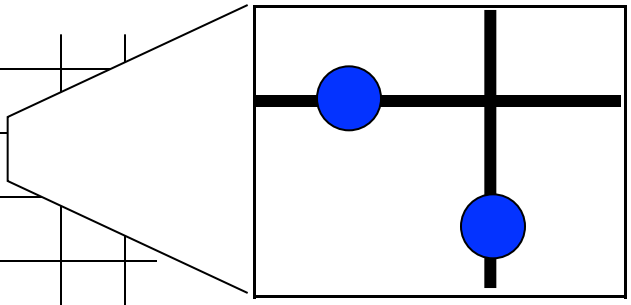
Kitaev

Physics, Physics, Physics, Physics!
Physics, Physics!

Kitaev, quant-ph/9707021, Annals Phys. 303, 2-30 (2003)

Kitaev's Model





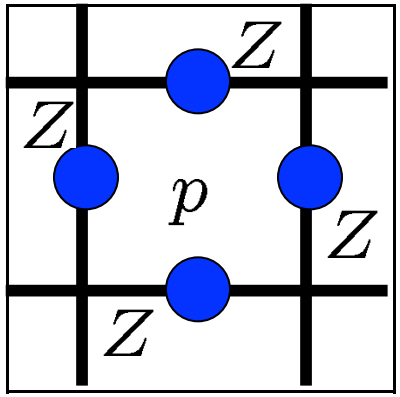
qubits on links

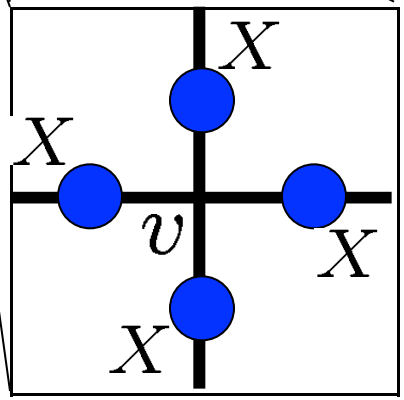
plaque operators:

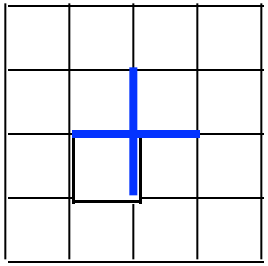
$$S_p = \prod_{e \in p} Z_e$$

vertex operators:

$$S_v = \prod_{e \in v} X_e$$







vertex and plaque operators all commute with each other

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \qquad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Kitaev, quant-ph/9707021

Kitaev's Model Hamiltonian

Hamiltonian

$$H = -J\left(\sum_p S_p + \sum_v S_v\right)$$

$$S_p|\psi\rangle = s_p|\psi\rangle \quad s_p \in \{-1, +1\}$$

$$S_v|\psi\rangle = s_v|\psi\rangle \quad s_v \in \{-1, +1\}$$

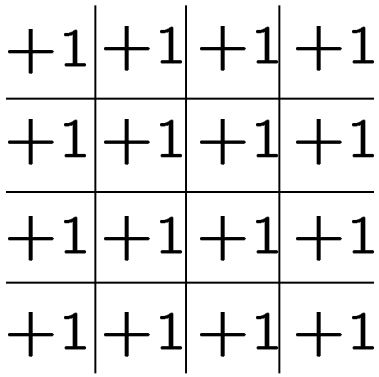
plaque operators:

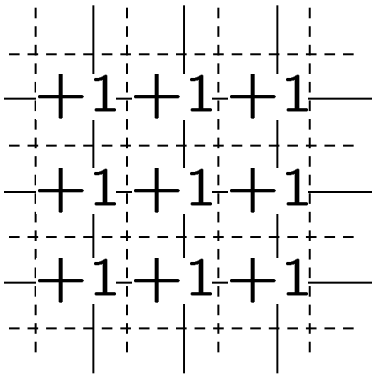
$$S_p = \prod_{e\in p} Z_e$$

vertex operators:

$$S_v = \prod_{e\in v} X_e$$

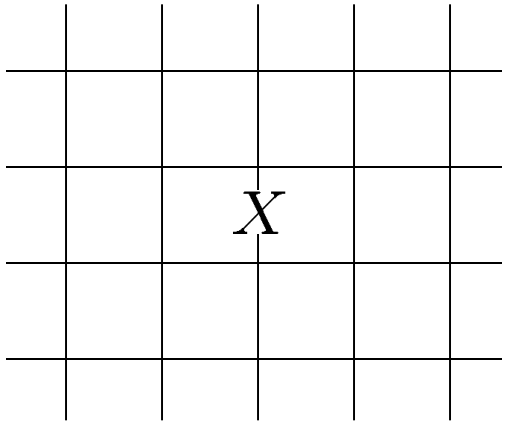
ground state has all $s_p = s_v = +1$





Kitaev, quant-ph/9707021

Kitaev's Model



plaque operators:

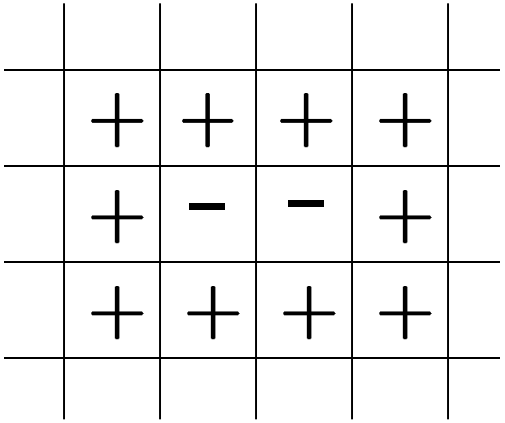
$$S_p = \prod_{e \in p} Z_e$$

vertex operators:

$$S_v = \prod_{e \in v} X_e$$

X anti-commutes with two plaque operators

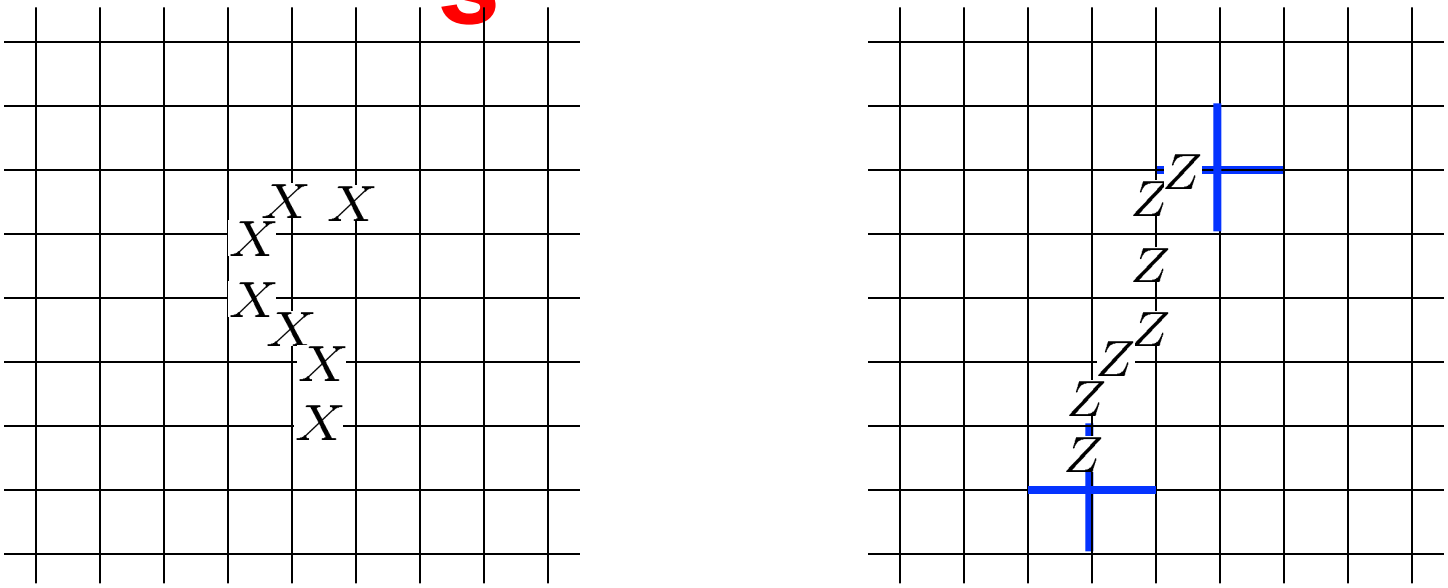
$$\{X, S_p\} = 0 \Rightarrow S_p(X|\psi_G\rangle) = -X|\psi_G\rangle$$



excitation is $4J$ above ground state

Kitaev, quant-ph/9707021

Excitations



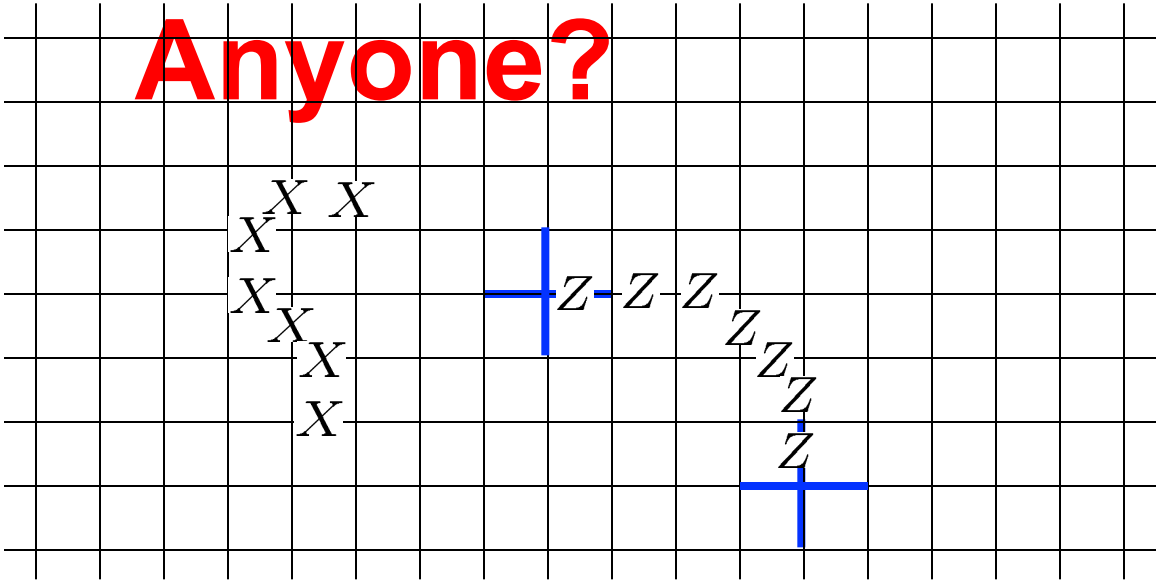
excitations particles come in pairs (particle/antiparticle) at end of "error" chains

two types of particles, X-type (live on vertices of dual lattice)

Z-type (live on vertices of the lattice)

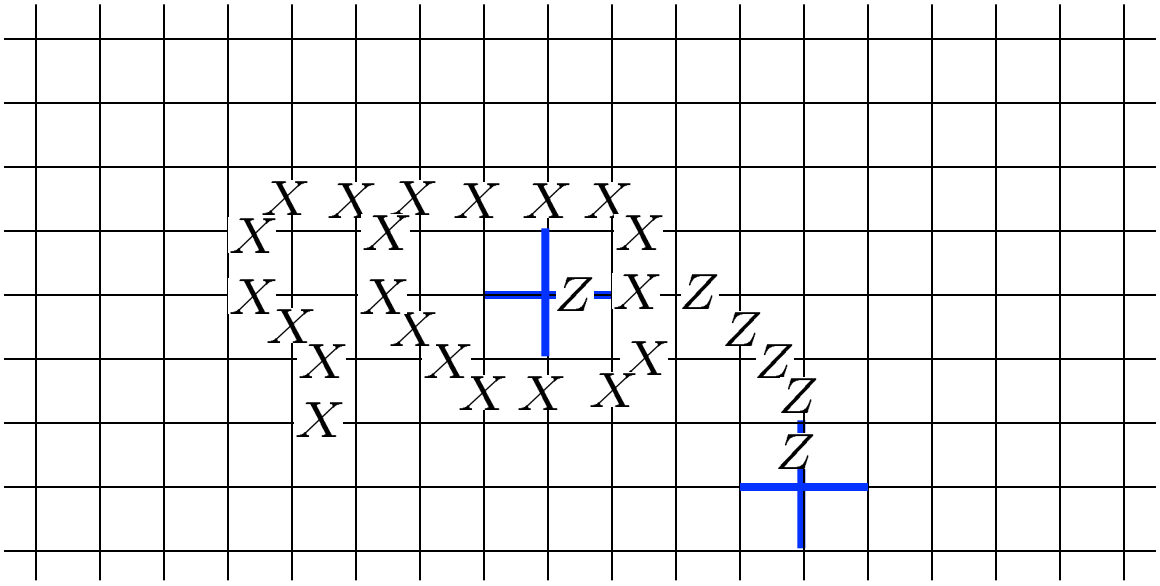
Kitaev, quant-ph/9707021

Anyons
Anyone?



Phase:

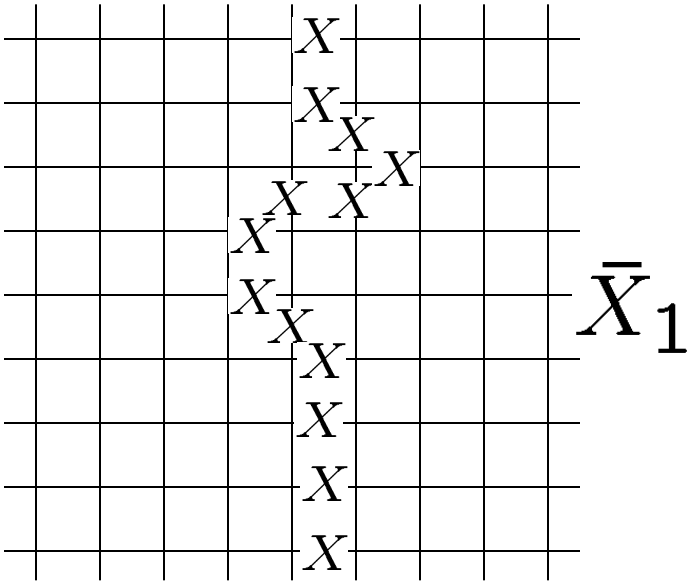
-1

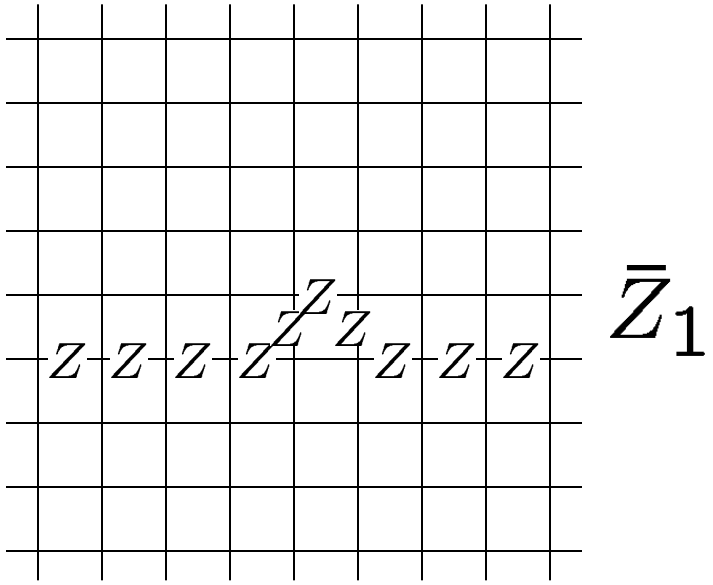


“relative semions”

Kitaev, quant-ph/9707021

A Qubit





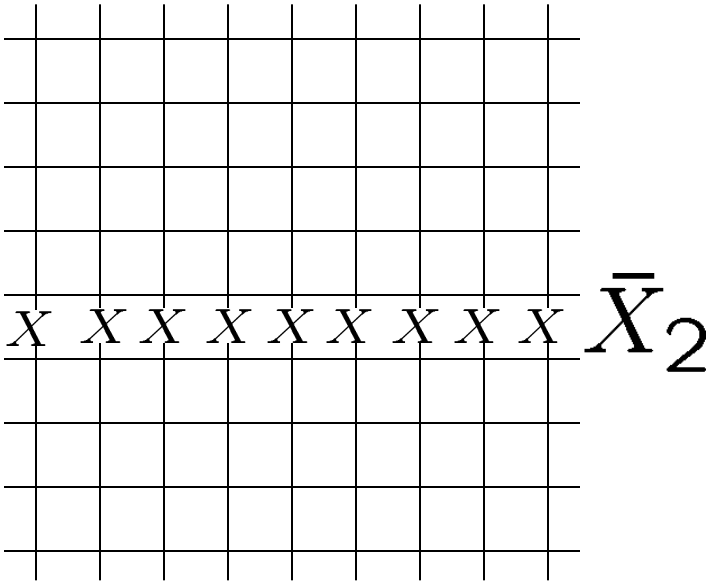
Moving an X or Z type particle on a homologically nontrivial path and annihilating with it's partner preserves the ground states

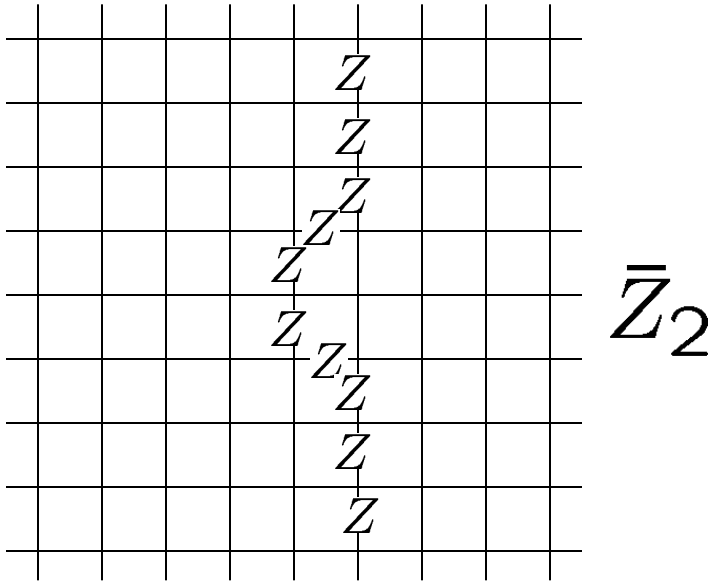
$$\bar{X}_1 \bar{Z}_1 \bar{X}_1 \bar{Z}_1 = -I$$

Action on ground state must be representation of non-Abelian group: Acts as encoded single qubit Pauli operator.

Kitaev, quant-ph/9707021

Two Qubits





Moving an X or Z type particle on a homologically nontrivial path and annihilating with it's partner preserves the ground states

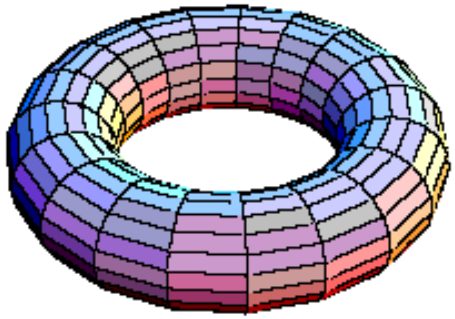
$$\bar{X}_2\bar{Z}_2\bar{X}_2\bar{Z}_2 = -I$$

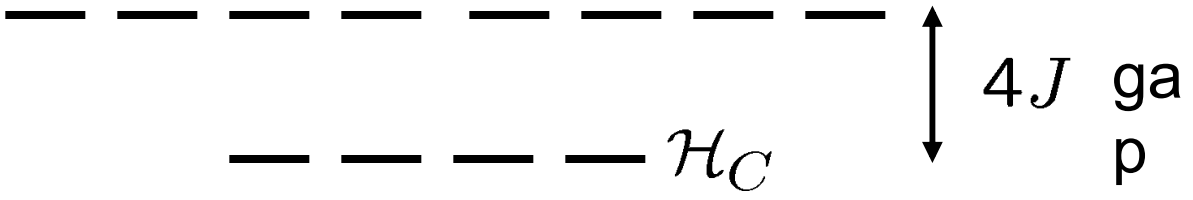
Action on ground state must be representation of non-Abelian group: Acts as encoded single qubit Pauli operator.

Kitaev, quant-ph/9707021

Topological P

Encode two qubits
into the ground state





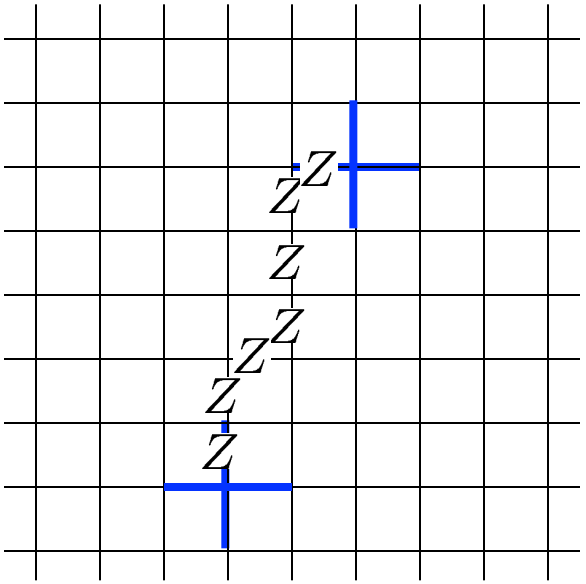
Perturbation theory:

$$V = \sum_v E_v + \sum_{vw} F_{vw}$$

But for $m \leq \left\lfloor \frac{l}{2} \right\rfloor$

$$\langle \phi_i | V^m | \phi_j \rangle = 0$$

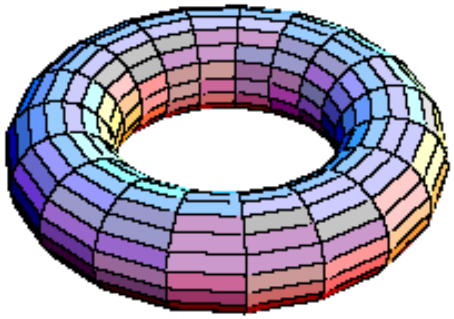
$$\langle \phi_i | V^m | \phi_i \rangle - \langle \phi_j | V^m | \phi_j \rangle = 0$$

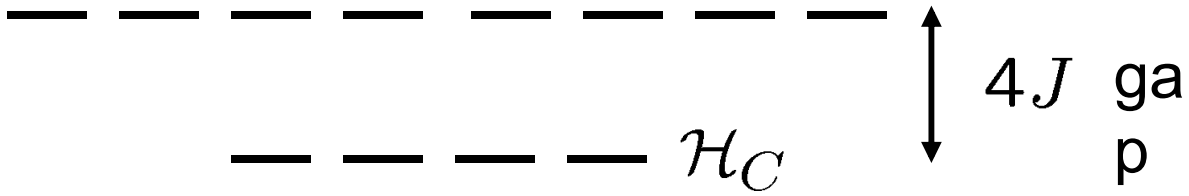


Kitaev, quant-ph/9707021

Topological Quantum Computing

Encode two qubits
into the ground state





If we keep $kT \ll 4J$ then quantum information encoded into ground state will be "topologically" robust to perturbation.

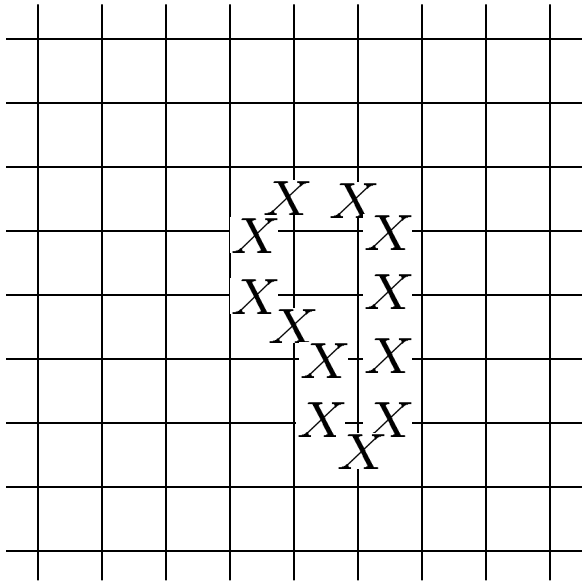
We can perform gates (in this example X and Z) by creating anyon pairs and moving them around the torus.

Question: can we do this manipulation fault-tolerantly?

Kitaev, quant-ph/9707021

Topological Error Correction

Different mode of using Kitaev's Model, as a component in a fault-tolerant quantum computer, encode into ground state



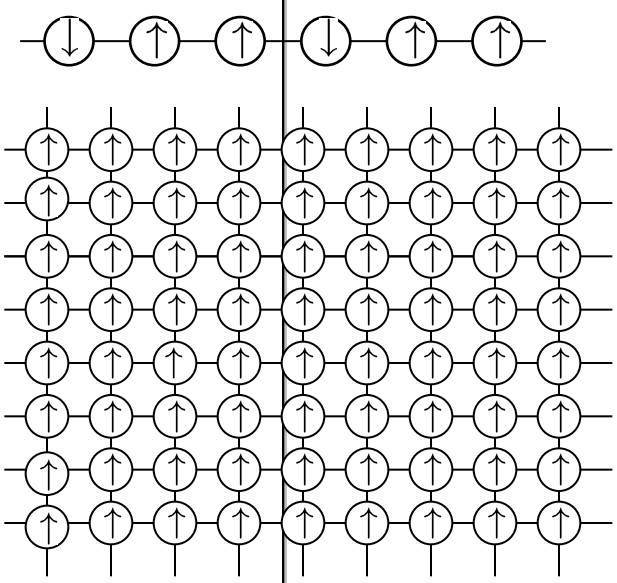
1. Errors occur
2. Measure vertex and plaque operators
3. Apply recovery operator
4. If created loop is homologically trivial, then we have restored information encoded into ground state.

Full analysis: measurements uncertain, recovery may fail, etc.
Threshold for fault-tolerant storage of quantum information:

Dennis, Kitaev, Landahl, Preskill, quant-ph/0110143, J. Math. Phys. 43, 4452 (2002)
Wang, Harrington, Preskill, quant-ph/0207088, Annals Phys. 303, 31 (2003)

Three Storage Testbeds

Ferromagnetic Ising models:



one dimensional

two (and >) dimensional

Toom's rule:

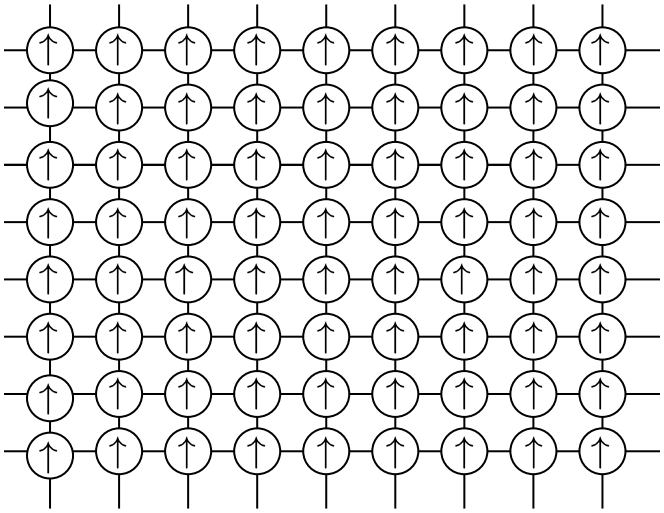
Toom, *Prob. of Inf. Trans.*, 10, 239 (1974)

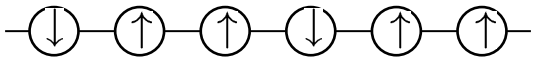
0	0	1	0	0	0	0	0
0	1	1	0	1	0	0	0
0	1	0	0	0	0	0	0
0	0	1	1	1	1	1	0
0	0	0	0	1	0	1	0
1	1	1	0	0	0	0	0

two dimensional noisy cellular automata

Ferromagnetic Ising Models







Ising model

spins on edges of a lattice: $s_i \in \{+1, -1\}$

energy of a configuration:

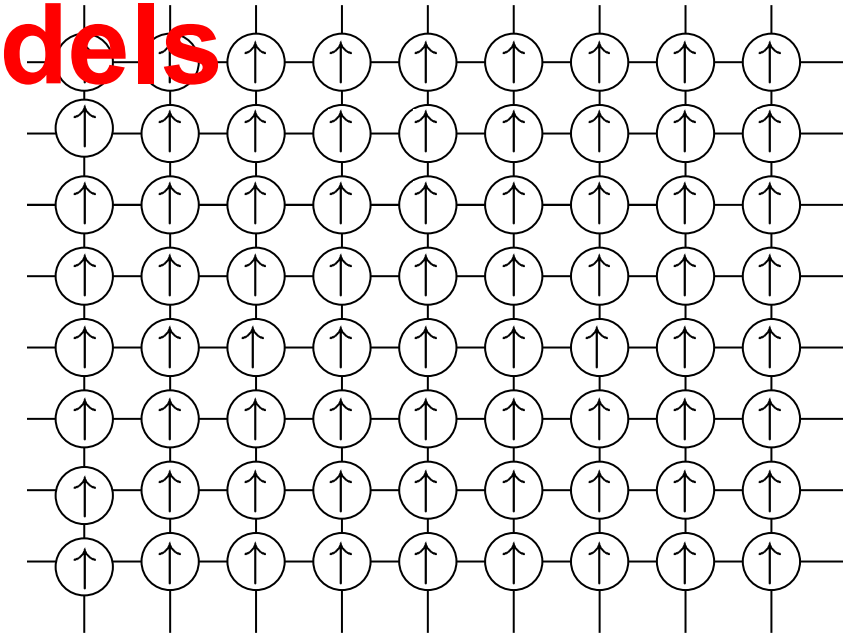
$$E = -J \sum_{\langle ij \rangle} s_i s_j + h \sum_i s_i \qquad J > 0$$

sum over neighbors

external magnetic field

$h = 0$ for now

Dynamic Ising Models



Dynamic Ising model (Metropolis update)

Interaction with environment causes spin flips

Update rule:

If flipping spin would decrease energy, flip spin

If flipping spin would increase energy, flip spin with probability $\exp(-\beta \Delta E)$

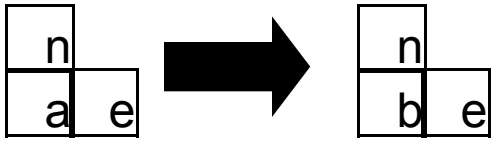
Toom's Rule



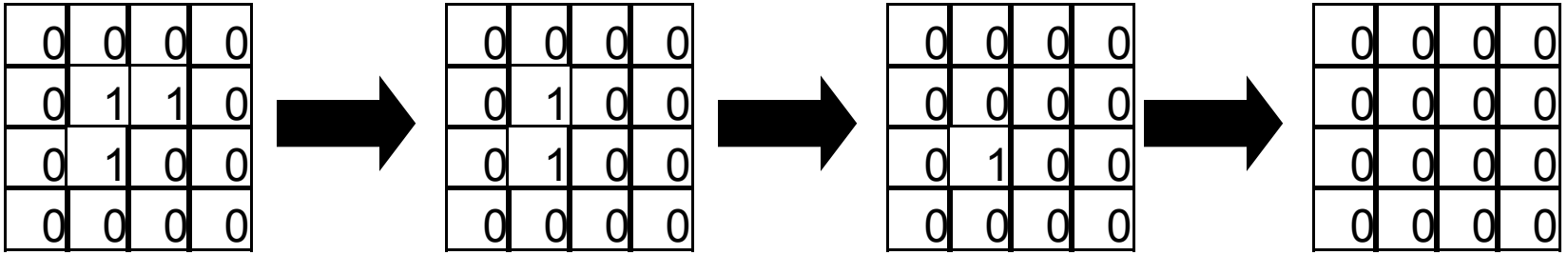
0	0	1	0	0	0	0	0
0	1	1	0	1	0	0	0
0	1	0	0	0	0	0	0
0	0	1	1	1	1	1	0
0	0	0	0	1	0	1	0
1	1	1	0	0	0	0	0

two dimensional noisy cellular automata

Update rule:



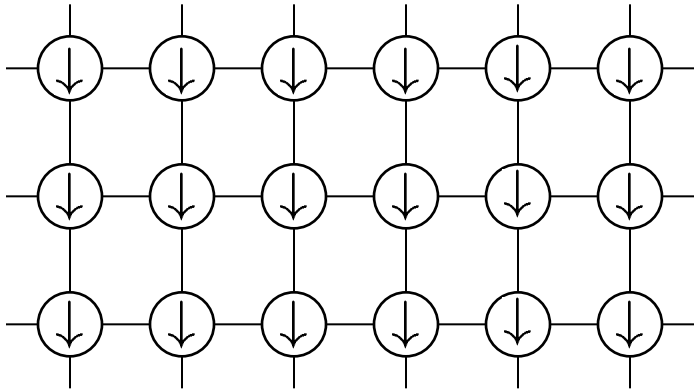
b=majority(n,e,b)



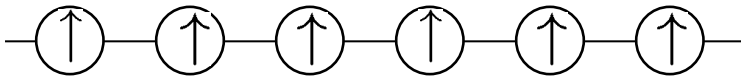
Update rule with noise: with some probability the update fails and a random state is assigned

Compare and Contrast

Encode information into the system and watch what happens



0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0



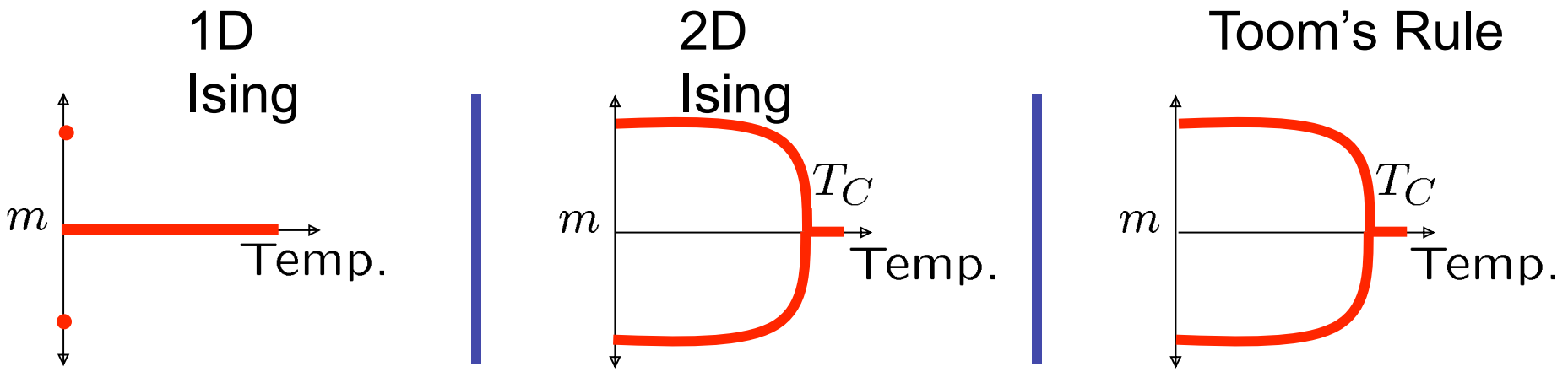
Redundancy code.

Order parameter is magnetization:

$$m = \frac{1}{N} \sum_i s_i$$

Storage

Long time behavior (not infinite time, but effectively)



Criticism: (1) thermodynamic limit taken

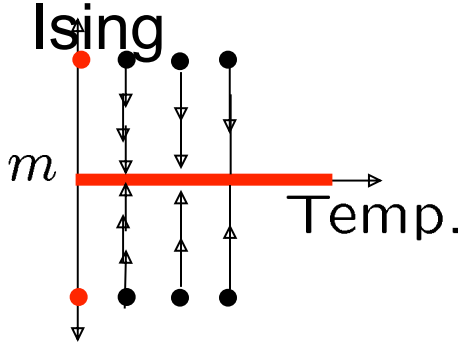
(2) what if relaxation to equilibrium takes a long time?

Storage



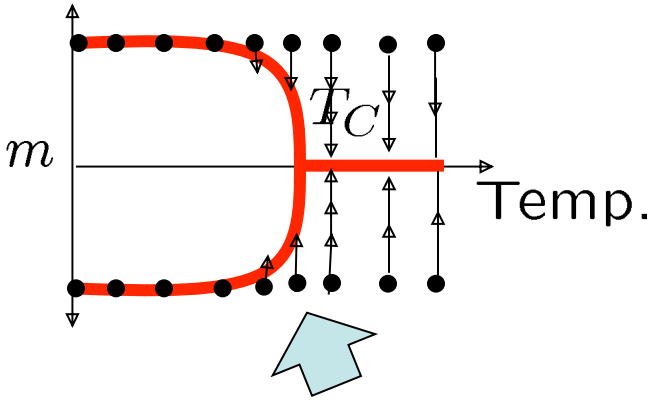
2nd attempt: 1D Ising

Relaxation to thermal



decay rate $\propto e^{-\beta J}$

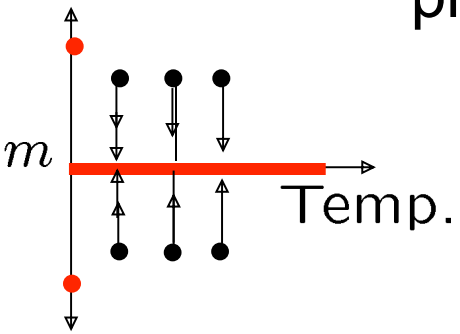
2D Ising, Toom's Rule

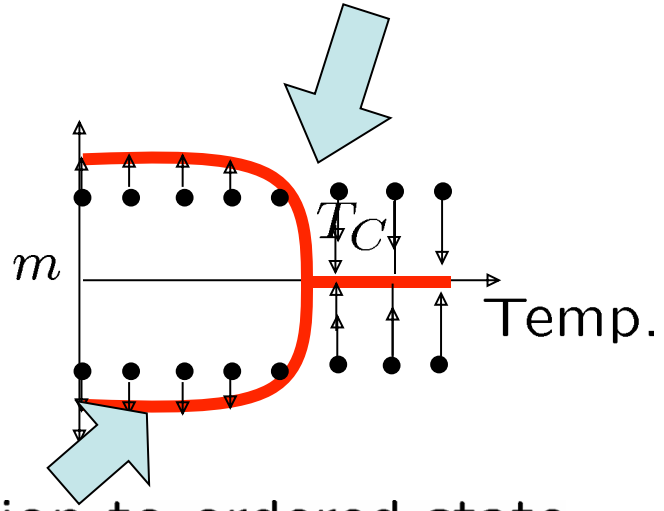


critical slowdown

$t_{th} \propto |T - T_c|^{z-\nu}$

Imperfect preparation





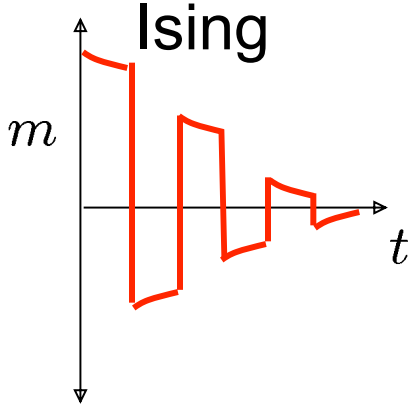
fast relaxation to ordered state

Manipulation



Flipping spins is imperfect

1D Ising

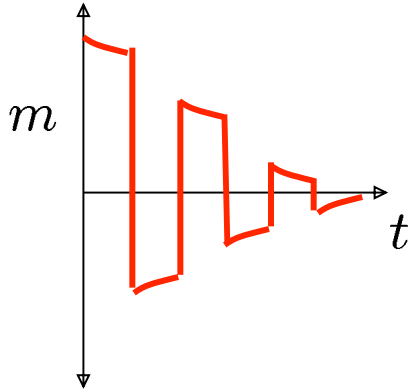


1D Ising
- temperature less than gap
- manipulation error must be small

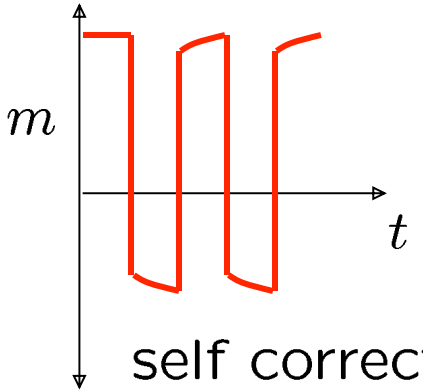
(i.e. NOT fault-tolerant)

2D Ising
- temperature less than critical temp

2D Ising
Toom's Rule



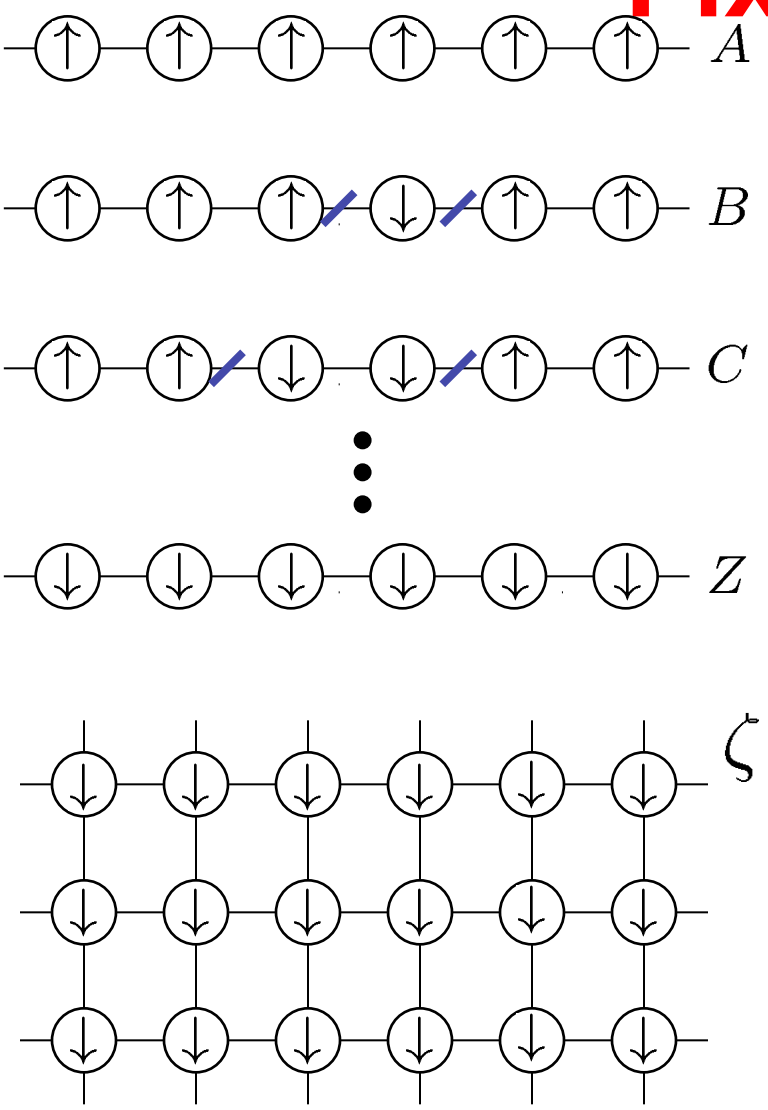
$T > T_C$

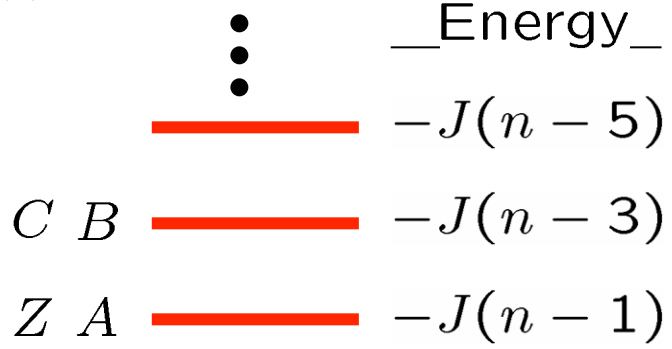


$T < T_C$

self correcting

“Self Fixing”

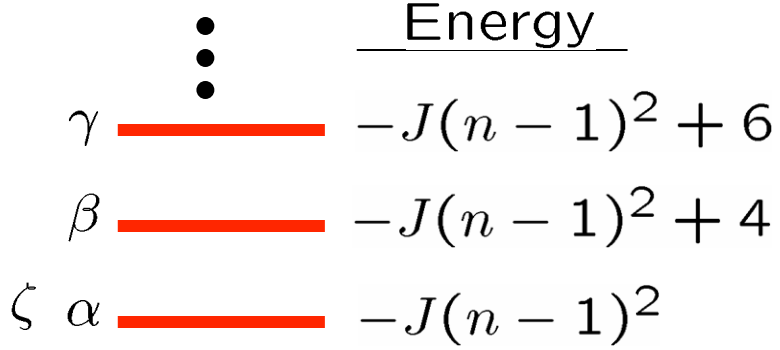




Excitations point-like

Energy to err is constant

Disordering entropy >> Energy



Excitations string-like

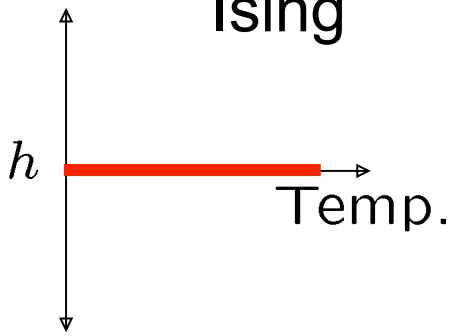
Energy to error is prop to size

Energy >> Disordering entropy

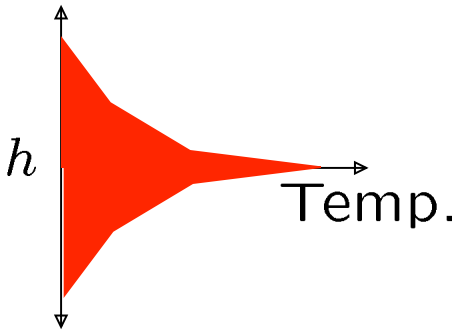
Toom's Rule Versus 2D Ising

Applied magnetic field (corresponds to a bias in noise)

2D Ising



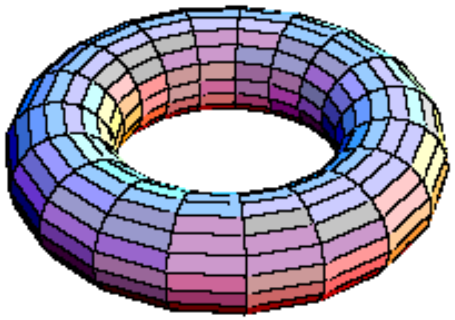
Toom's Rule



Regions with two coexisting phases

External magnetic fields disorder magnetic media

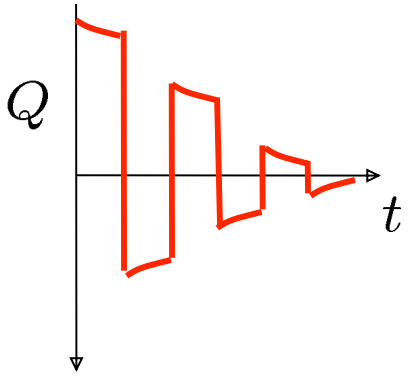
Whence Kitaev's Model?

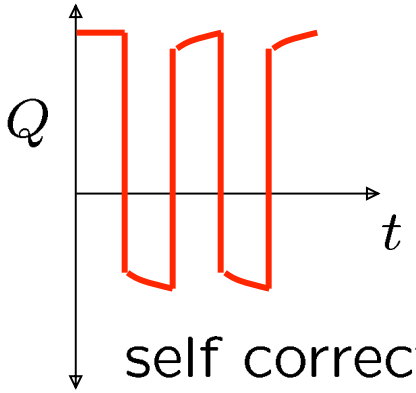


$$H = -J\left(\sum_p S_p + \sum_v S_v\right)$$

Robust storage of quantum information.

ut does it provide a mechanism for
obust **manipulation** of quantum
nformation?

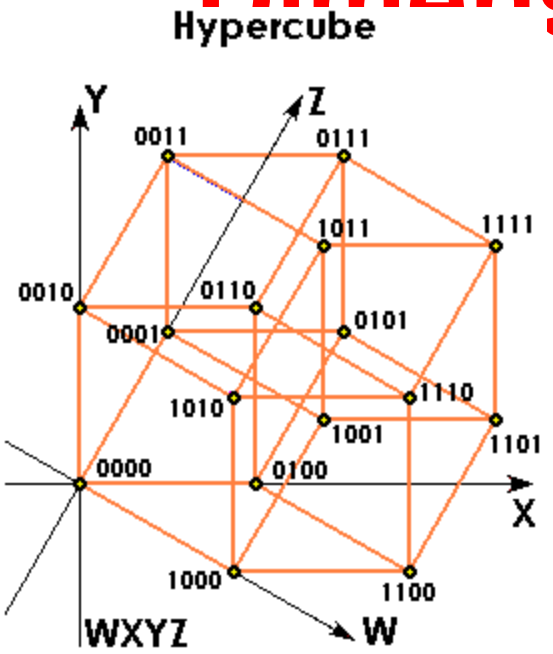




$T < T_C$

Is it possible to construct a Hamiltonian, which allows for fault-tolerant manipulation as well?

Into the Fourth Dimension



Toric code in four dimension

$$H = -J\left(\sum_p S_p + \sum_v S_v\right)$$

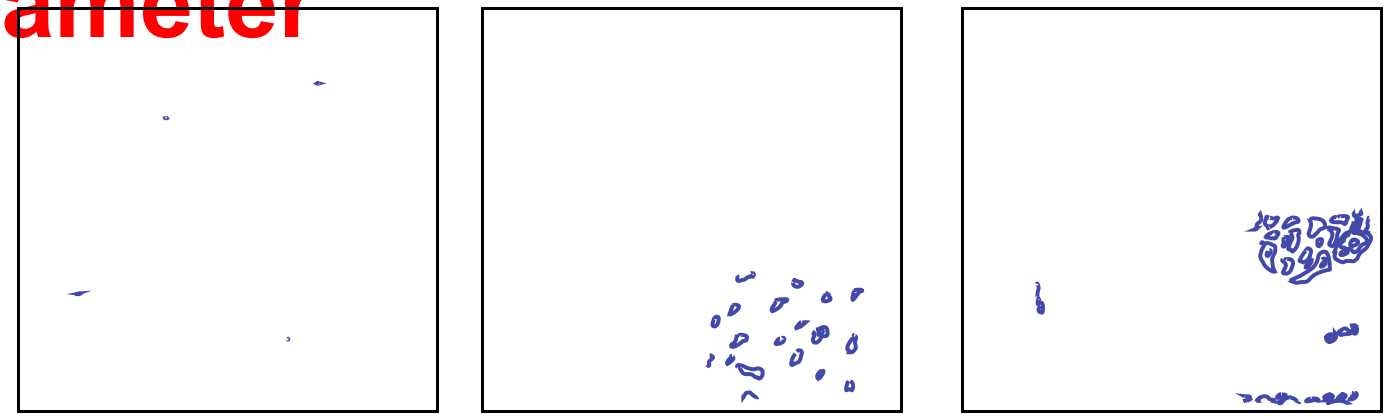
6-local operators

Excitations require energy proportional to perimeter of erred region.

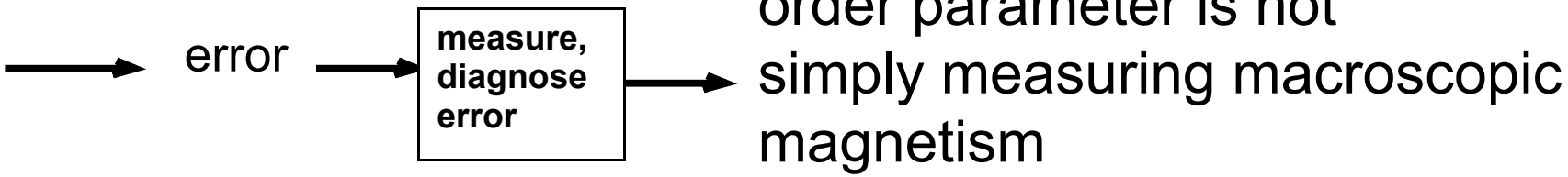
Major open question: are there three dimensional systems which possess a similar stability to quantum errors?

Dennis, Kitaev, Landahl, Preskill, quant-ph/0110143, J. Math. Phys. 43, 4452 (2002)

Quantum Correcting Order Parameter

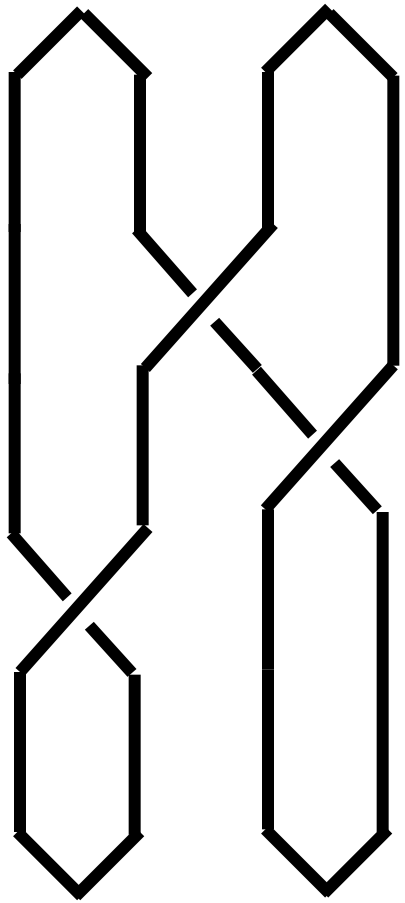


$T \approx 0$ $T \approx T_C$



Topological Quantum Computation

"Large" class of physical theories have non-abelian anyons.



3. Fusing of anyons ⇔ Measurement

2. Braiding of anyons ⇔ Unitary gates

1. Creation of anyons □⇔ Preparation

Topological Developments

Possible phases of fractional quantum Hall effect

- v=5/2 may have non-abelian anyons
  not universal by just braiding
  (See however Bravyi PRA 73, 042313 (2003))
- v=12/5 may have non-abelian anyons, universal

Plethora of other systems, some exactly solvable, proposed to have topological phases and be universal [Search word: Freedman, Nayak, Kitaev]

Color codes [see talk by Héctor Bombin]

Implementation ideas in optical lattices and superconducting circuits [see talk by Jiannis Pachos and Gavin Brennen]

Joke Deleted By Hollywood Censors

Small Scale Memories

Are there small simple realistically implementable versions of Kitaev's idea of using an energy gap to protect quantum information?

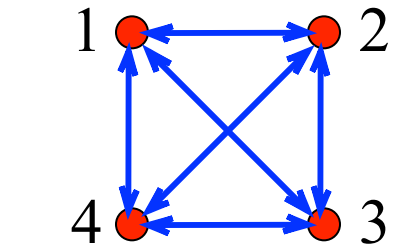
Supercoherent Qubits

D. Bacon, K.R. Brown, and K.B. Whaley, *Phys. Rev. Lett.* 87, 247902 (2001)
Y.S. Weinstein and C. S. Hellberg, *Phys. Rev. Lett.* 98, 110501 (2007)

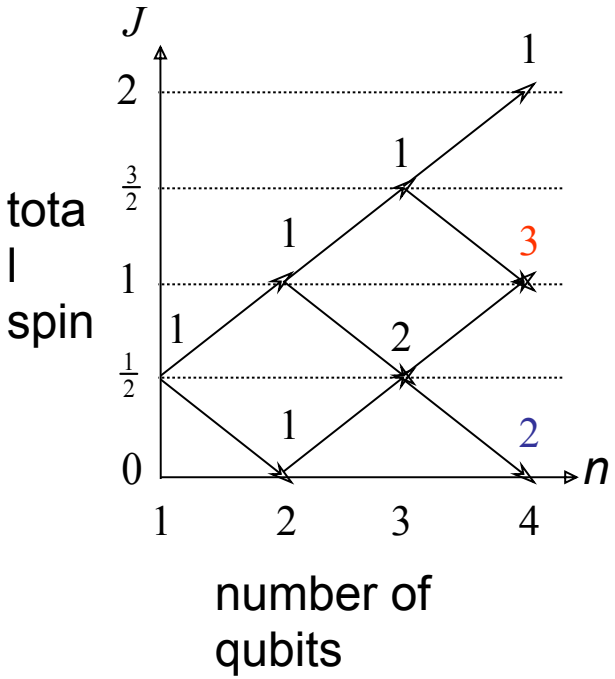
$$\mathbf{H} = \Delta \sum_{i>j=1}^{4} \hat{\mathbf{s}}_i \cdot \hat{\mathbf{s}}_j = \frac{\Delta}{2}\left(\hat{\mathbf{S}}^2 - 3\mathbf{I}\right)$$

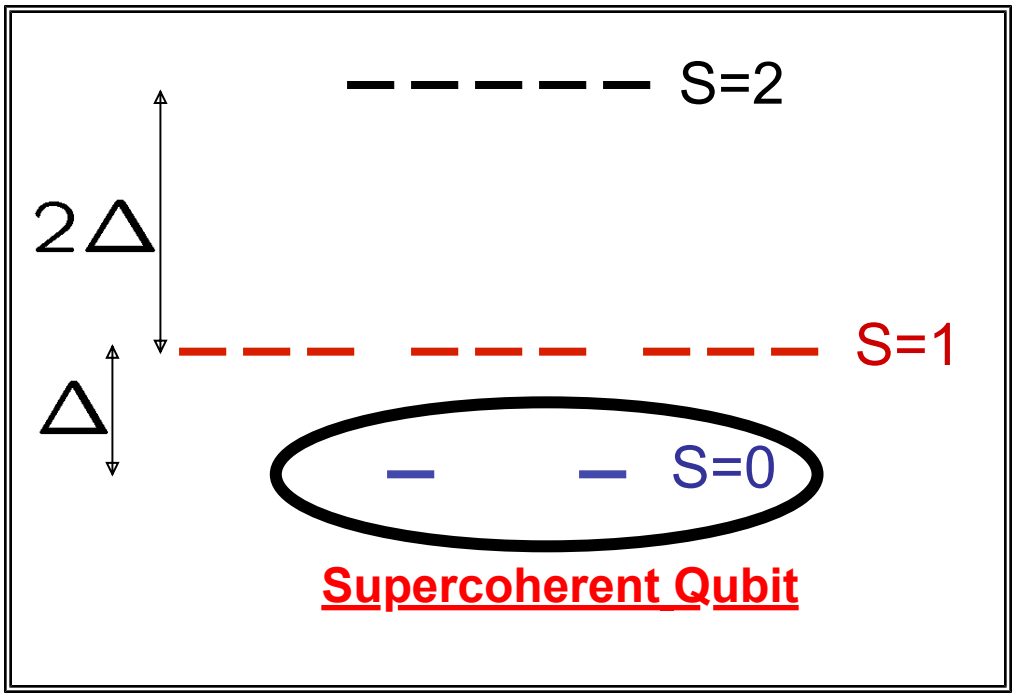
total spin

$$\hat{\mathbf{s}}_i = (\mathbf{s}_x, \mathbf{s}_y, \mathbf{s}_z)|_{i\text{th qubit}} \qquad \hat{\mathbf{S}} = \sum_{i=1}^{4} \hat{\mathbf{s}}_i$$

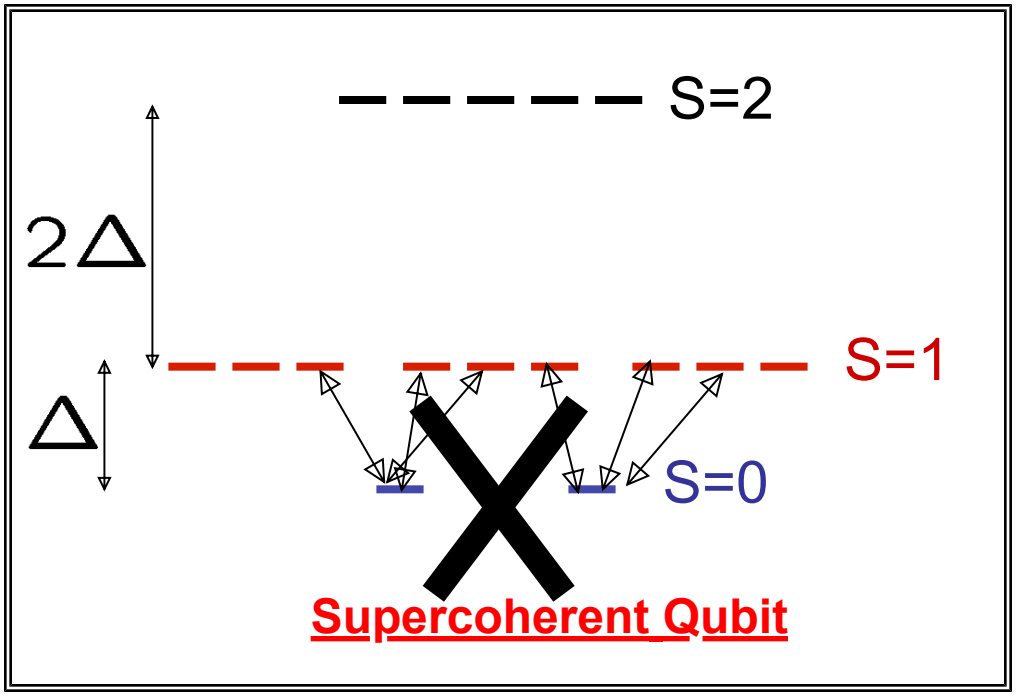


= exchange interaction



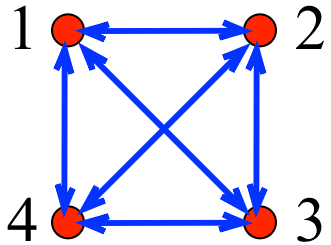


Run Uphill



“heating”

$(s_1)_+$



$(s_3)_z$

“nondissipative”

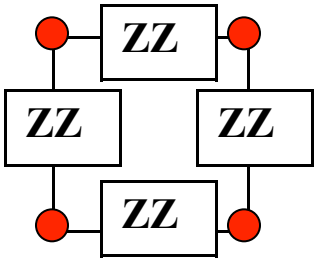
All single qubit operators $(s_i)_\alpha$ take S=0 states to S>0 states

Supercoherence:

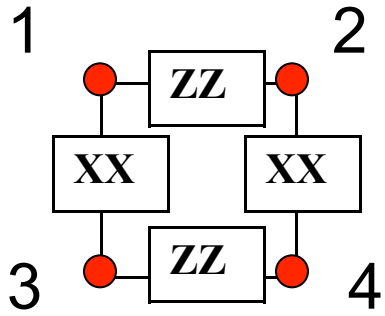
made all single qubit processes “heating”!

At kT< Δ single qubit decoherence mechanisms suppressed

Another Small Quantum Try



classical



quantum

Hamiltonian

$$H = -(Z_1Z_2 + Z_3Z_4 + X_1X_3 + X_2X_4)$$

quantum error detecting code

$$S_1 = Z_1Z_2Z_3Z_4$$
$$S_2 = X_1X_2X_3X_4$$

symmetry

$$[H, S_1] = [H, S_2] = 0$$

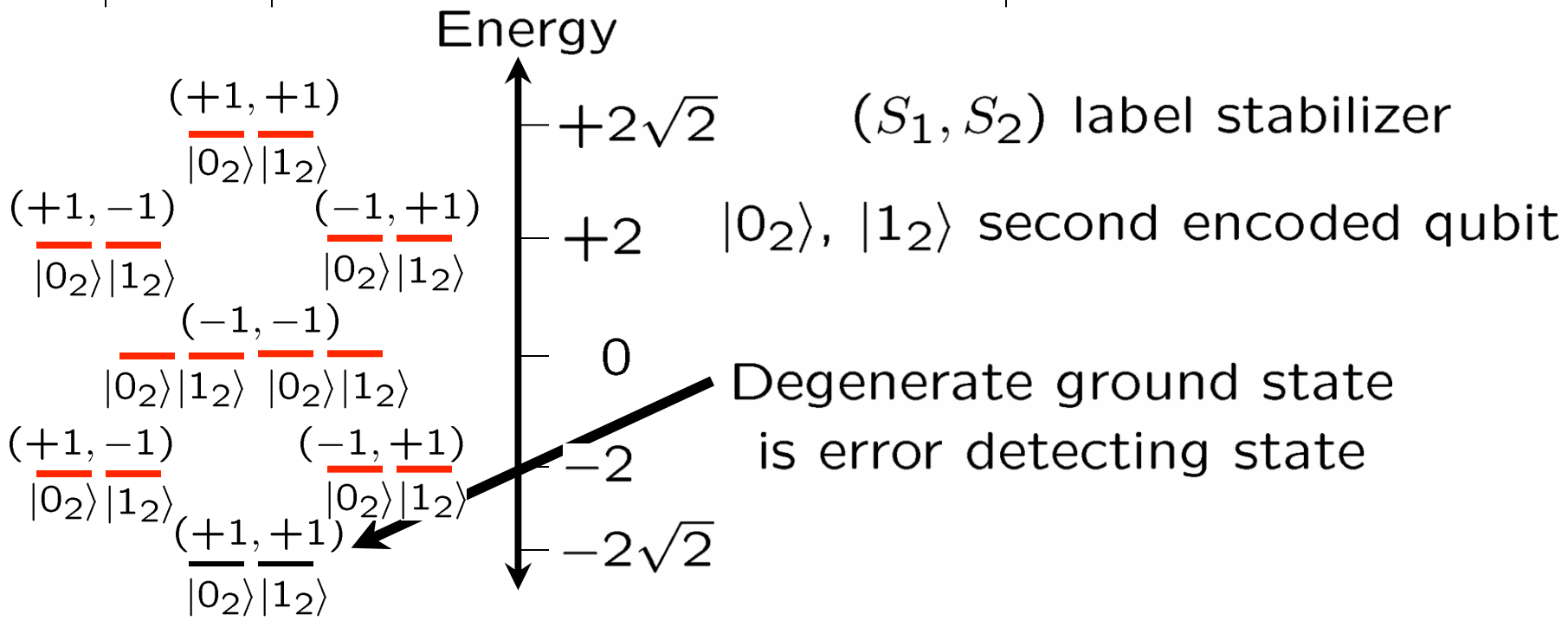
$$\bar{Z}_1 = Z_1Z_2 \qquad \bar{X}_1 = Z_1Z_3$$
$$\bar{Z}_2 = Z_1Z_3 \qquad \bar{X}_2 = X_1X_2$$

$$H = -\bar{Z}_1(I + S_1) - \bar{X}_1(I + S_2)$$

simultaneously diagonalize with symmetries

H does not depend on second qubit

S_1	S_2	$H = \bar{X}_1(1+S_1) + \bar{Z}_1(1+S_2)$	Eigenvalues
$+1$	$+1$	$2(\bar{X}_1 + \bar{Z}_1)$	$\pm 2\sqrt{2}$
$+1$	-1	$2\bar{X}_1$	± 2
-1	$+1$	$2\bar{Z}_1$	± 2
-1	-1	0	0



Supercoherence

$$\mathbf{H} = \mathbf{X}_1(1 + S_1) + \mathbf{Z}_1(1 + S_2)$$

$\mathbf{S}_i\mathbf{P} = -\mathbf{P}\mathbf{S}_i$ for some i $\quad$ $\mathbf{P}$ single qubit operator

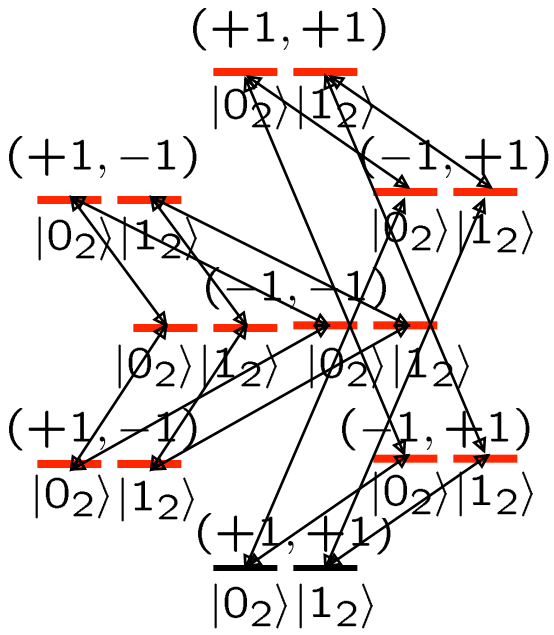
ALL single qubit errors take degenerate ground state to higher energy levels.

Single qubit errors change value of (S1,S2) and hence take ground state to higher energy level.

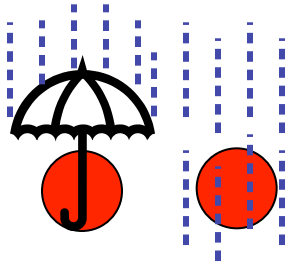
$$\underline{(+1,+1)}$$
$$|0_2\rangle \qquad |1_2\rangle$$

✗ **FORBIDDEN**

Effect of **ZIII**



Subsystem Error Correction



Only need to protect subsystem,
not full subspaces.

$$\mathcal{H}_{sys} = (\mathcal{C} \otimes \mathcal{D}) \oplus \mathcal{P}$$

encoding subsystem

$$(|\phi\rangle\langle\phi| \otimes \rho_D) \oplus 0 \;\; \mathcal{E} \;\; \mathcal{R} \;\; (|\phi\rangle\langle\phi| \otimes \tilde{\rho}_D) \oplus 0$$

error

recover y

encoded quantum information

recovery need not perfectly restore system, but only preserve information encoded into subsystem.

Kribbs, Laflamme, Poulin, PRL 94, 180501 (2005)
D. Poulin, PRL 95, 230504, (2005)

A Silly Stabilizer Code

Stabilizer generators:

$$S_1 = \begin{matrix} X & X & X \\ X & X & X \\ I & I & I \end{matrix} \quad S_2 = \begin{matrix} I & I & I \\ X & X & X \\ X & X & X \end{matrix} \quad S_3 = \begin{matrix} Z & Z & I \\ Z & Z & I \\ Z & Z & I \end{matrix} \quad S_4 = \begin{matrix} I & Z & Z \\ I & Z & Z \\ I & Z & Z \end{matrix}$$

Logical operators:

$$\bar{X}_1 = \begin{matrix} X & I & I \\ X & I & I \\ I & I & I \end{matrix} \quad \bar{X}_2 = \begin{matrix} I & I & X \\ I & I & X \\ I & I & I \end{matrix} \quad \bar{X}_3 = \begin{matrix} I & I & I \\ X & I & I \\ X & I & I \end{matrix} \quad \bar{X}_4 = \begin{matrix} I & I & I \\ I & I & X \\ I & I & X \end{matrix}$$

$$\bar{Z}_1 = \begin{matrix} Z & Z & I \\ I & I & I \\ I & I & I \end{matrix} \quad \bar{Z}_2 = \begin{matrix} I & Z & Z \\ I & I & I \\ I & I & I \end{matrix} \quad \bar{Z}_3 = \begin{matrix} I & I & I \\ I & I & I \\ Z & Z & I \end{matrix} \quad \bar{Z}_4 = \begin{matrix} I & I & I \\ I & I & I \\ I & Z & Z \end{matrix}$$

$$\bar{X}_5 = \begin{matrix} X & X & X \\ X & X & X \\ X & X & X \end{matrix} \quad \bar{Z}_5 = \begin{matrix} Z & Z & Z \\ Z & Z & Z \\ Z & Z & Z \end{matrix}$$

Bacon, Phys. Rev A, 73, 012340

Subsystem Encoding

$$S_1 = \begin{matrix} X & X & X \\ X & X & X \\ I & I & I \end{matrix} \quad S_2 = \begin{matrix} I & I & I \\ X & X & X \\ X & X & X \end{matrix} \quad S_3 = \begin{matrix} Z & Z & I \\ Z & Z & I \\ Z & Z & I \end{matrix} \quad S_4 = \begin{matrix} I & Z & Z \\ I & Z & Z \\ I & Z & Z \end{matrix}$$

$$\mathcal{H} = \bigoplus_{S_1,S_2,S_3,S_4 \in \{\pm 1\}} \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \mathcal{H}_3 \otimes \mathcal{H}_4 \otimes \mathcal{H}_5$$

$$\bar{X}_5 = \begin{matrix} X & X & X \\ X & X & X \\ X & X & X \end{matrix} \quad \bar{Z}_5 = \begin{matrix} Z & Z & Z \\ Z & Z & Z \\ Z & Z & Z \end{matrix}$$

Sample Error

$$\begin{matrix} I & I & I \\ I & Z & I \\ I & I & I \end{matrix}$$



$$\begin{matrix} X & X & X \\ X & X & X \\ I & I & I \end{matrix} \qquad \begin{matrix} I & I & I \\ X & X & X \\ X & X & X \end{matrix} \qquad \begin{matrix} Z & Z & I \\ Z & Z & I \\ Z & Z & I \end{matrix} \qquad \begin{matrix} I & Z & Z \\ I & Z & Z \\ I & Z & Z \end{matrix}$$

$$\begin{matrix} I & I & I \\ I & Z & I \\ I & I & I \end{matrix}$$



$$\begin{matrix} X & X & X \\ X & X & X \\ I & I & I \end{matrix} \qquad \begin{matrix} I & I & I \\ X & X & X \\ X & X & X \end{matrix} \qquad \begin{matrix} Z & Z & I \\ Z & Z & I \\ Z & Z & I \end{matrix} \qquad \begin{matrix} I & Z & Z \\ I & Z & Z \\ I & Z & Z \end{matrix}$$

Syndrome

$\begin{matrix} I & I & I \\ Z & I & I \\ I & I & I \end{matrix}$	$\begin{matrix} I & I & I \\ I & Z & I \\ I & I & I \end{matrix}$	$\begin{matrix} I & I & I \\ I & I & Z \\ I & I & I \end{matrix}$
$\begin{matrix} I & I & I \\ Z & Z & I \\ I & I & I \end{matrix}$	$\begin{matrix} I & I & I \\ I & I & I \\ I & I & I \end{matrix}$	$\begin{matrix} I & I & I \\ I & Z & Z \\ I & I & I \end{matrix}$
$\bar{Z}_1 \bar{Z}_3 S_3$	I	$\bar{Z}_2 \bar{Z}_4 S_4$

Fix

Resulting Operation

Subsystem Code

Stabilizer generators:

$$\begin{matrix} Z & Z & I & I & Z & Z \\ Z & Z & I & I & Z & Z \\ Z & Z & I & I & Z & Z \end{matrix}$$

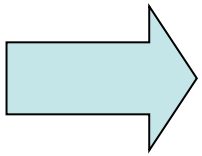
plus encoded Z operators

$$\bar{Z}_1 = \begin{matrix} Z & Z & I \\ I & I & I \\ I & I & I \end{matrix} \quad \bar{Z}_2 = \begin{matrix} I & Z & Z \\ I & I & I \\ I & I & I \end{matrix} \quad \bar{Z}_3 = \begin{matrix} I & I & I \\ I & I & I \\ Z & Z & I \end{matrix} \quad \bar{Z}_4 = \begin{matrix} I & I & I \\ I & I & I \\ I & Z & Z \end{matrix}$$

generate all even number of Z's per row

Measuring +1,-1 eigenvalue of

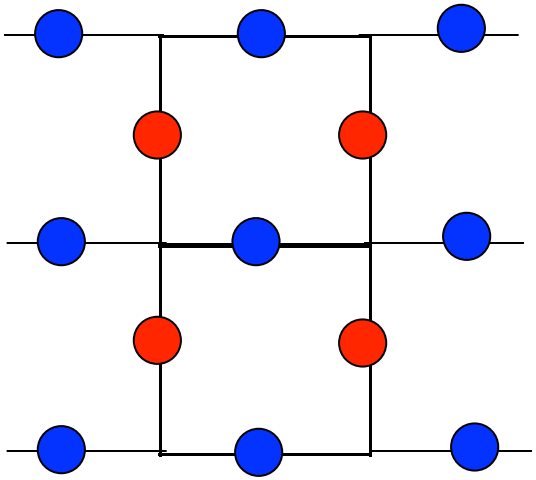
$$\begin{matrix} X & X & X & I & I & I \\ X & X & X & X & X & X \\ I & I & I & X & X & X \end{matrix}$$



Fix modulo this group i.e. do nothing to 5th encoded qubit!

[Bacon, Phys. Rev A, 73, 012340, 2006]

Relation to Other Codes



Aliferis and Cross, Phys. Rev. Lett. 98 220502 (2007)

Measure red qubits in computational basis leaves blue qubits in subsystem code.

Code is closely related to Shor's original 9 qubit code:

$$S_1 = \begin{matrix} X & X & X \\ X & X & X \\ I & I & I \end{matrix} \quad S_2 = \begin{matrix} I & I & I \\ X & X & X \\ X & X & X \end{matrix} \quad S_3 = \begin{matrix} Z & Z & I \\ I & I & I \\ I & I & I \end{matrix} \quad S_4 = \begin{matrix} I & Z & Z \\ I & I & I \\ I & I & I \end{matrix}$$

$$S_5 = \begin{matrix} I & I & I \\ I & I & I \\ Z & Z & I \end{matrix} \quad S_6 = \begin{matrix} I & I & I \\ I & I & I \\ I & Z & Z \end{matrix} \quad S_7 = \begin{matrix} I & I & I \\ Z & Z & I \\ I & I & I \end{matrix} \quad S_8 = \begin{matrix} I & I & I \\ I & Z & Z \\ I & I & I \end{matrix}$$

Generalized Shor Subsystem Codes

Two classical codes: C1=[n1,k1,d1] and C2= [n2,k2,d2]

Quantum Shor subspace code
[[n1n2,k1k2,min(d1,d2)]]
Number of stabilizers = (n1-k1)n2+(n2-k2)

Quantum Shor subsystem code
[[n1n2,k1k2,min(d1,d2)]]
Number of stabilizers= (n1-k1)k2+(n2-k2)k1

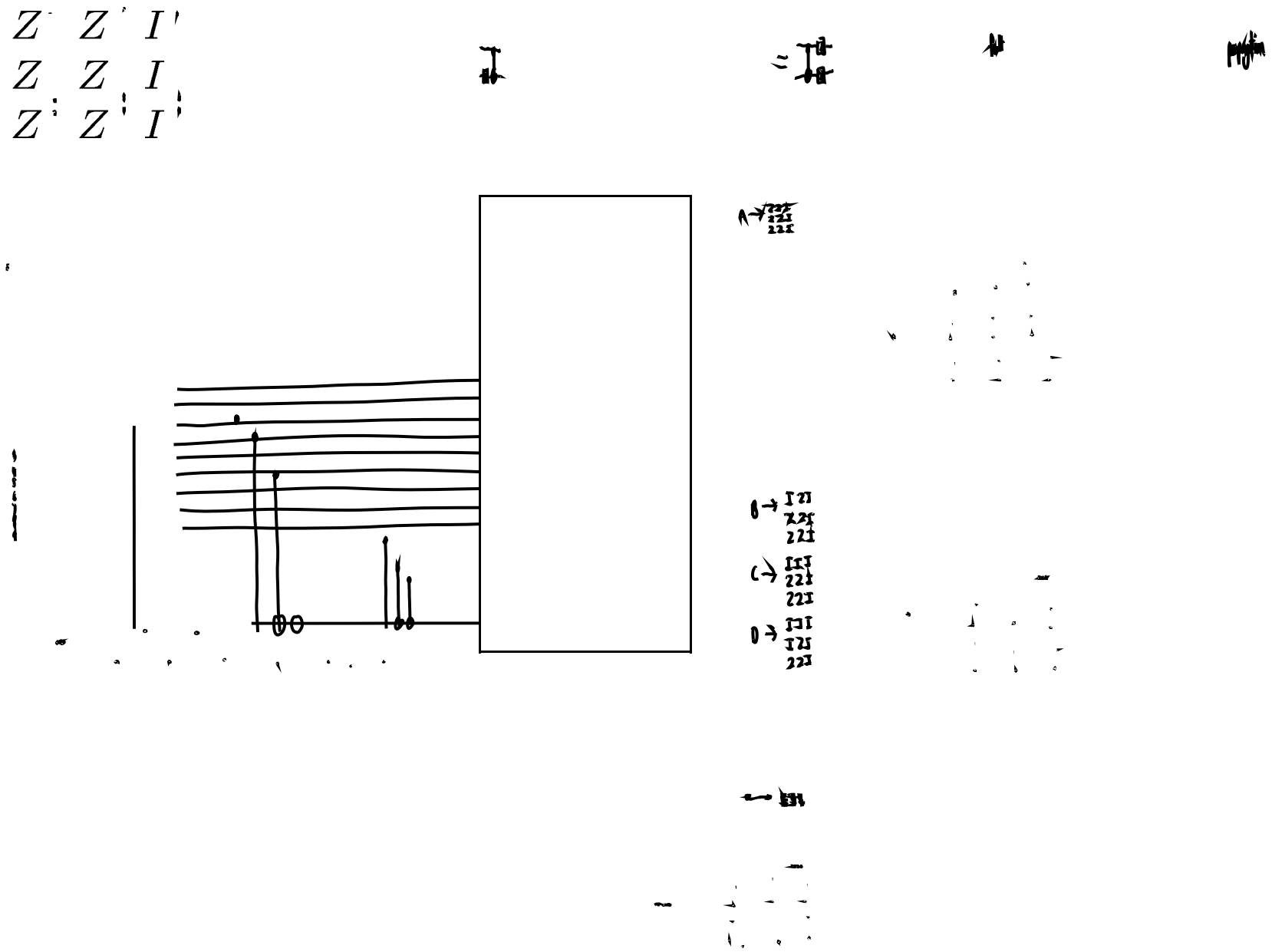
Bacon and Casaccnio, quant-ph/0610088 (2006)

More Subsystem Codes

Aly, Klappenecker, Sarvepalli, quant-ph/0610153 (2006)
Klappenecker, Sarvepalli, quant-ph/0703213 (2007)

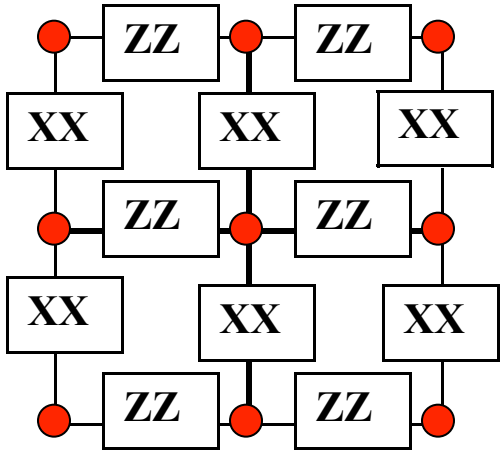
[see talk by Salah Aly]

Extra Fault-Tolerant Properties



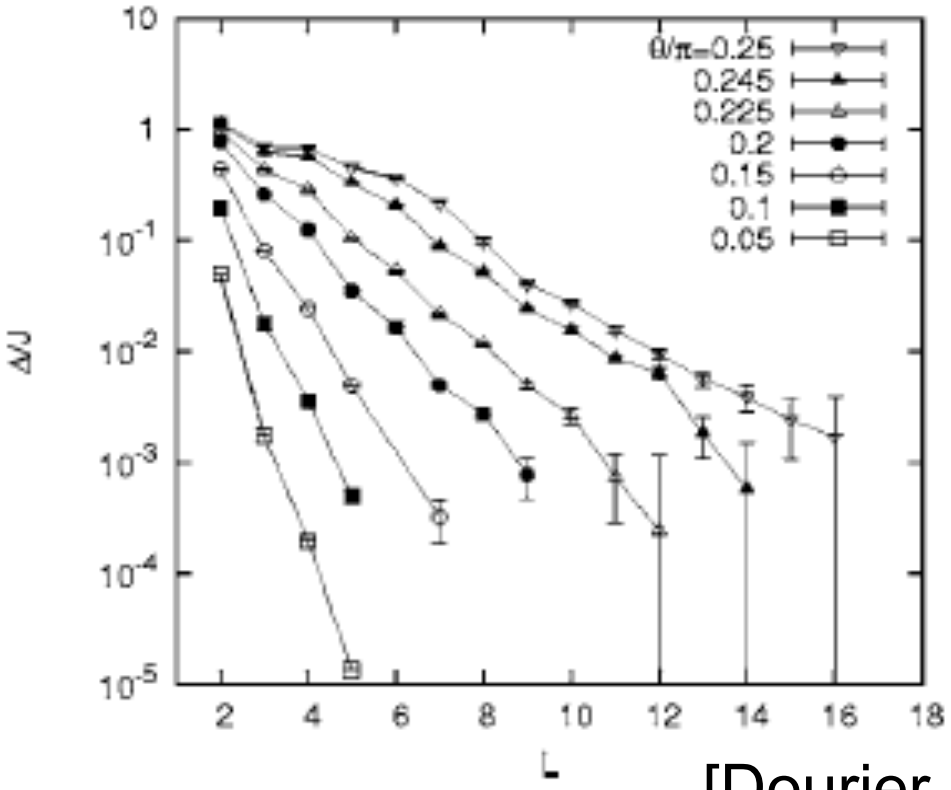
Quantum Compass Model

$$H = -J_X \sum_{ij} X_{i,j} X_{i+1,j} - J_Z \sum_{ij} Z_{i,j} Z_{i,j+1}$$



Ground state is subsystem code: does it have an energy gap?

Problems:



[Dourier, Becca, Mila 05]

$$H = -J \sum_{i,j=1}^{n} \cos(\theta) X_{i,j} X_{i+1,j} + \sin(\theta) Z_{i,j} Z_{i,j+1}$$

no gap in the thermodynamic limit

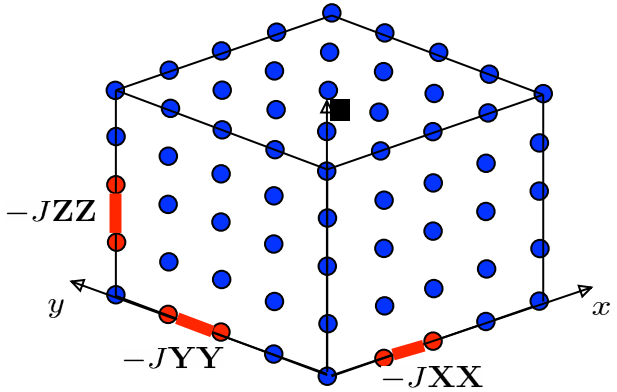
Long Range Quantum Compass Model?

$$H = -\sum_{i,j=1}^{n}\sum_{k=1}^{n-1} J_X(k) X_{i,j} X_{i+k,j} + J_Z(k) Z_{i,j} Z_{i,j+k}$$

Proposed implementation in ion traps (See however Brown 0705.2370)

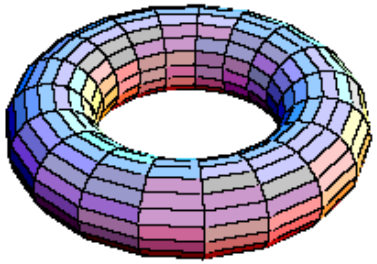
Milman, Maineult, et al Phys. Rev. Lett. 99, 020503 (2007).

3D Codes Quantum Compass?



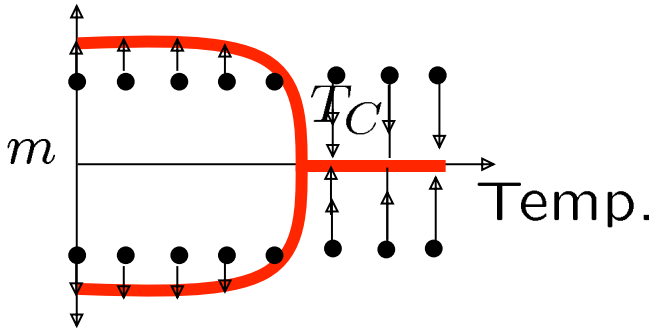
In ect

- Toric code / ion

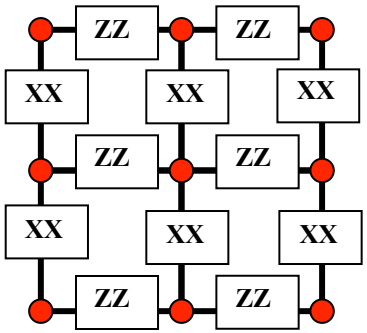




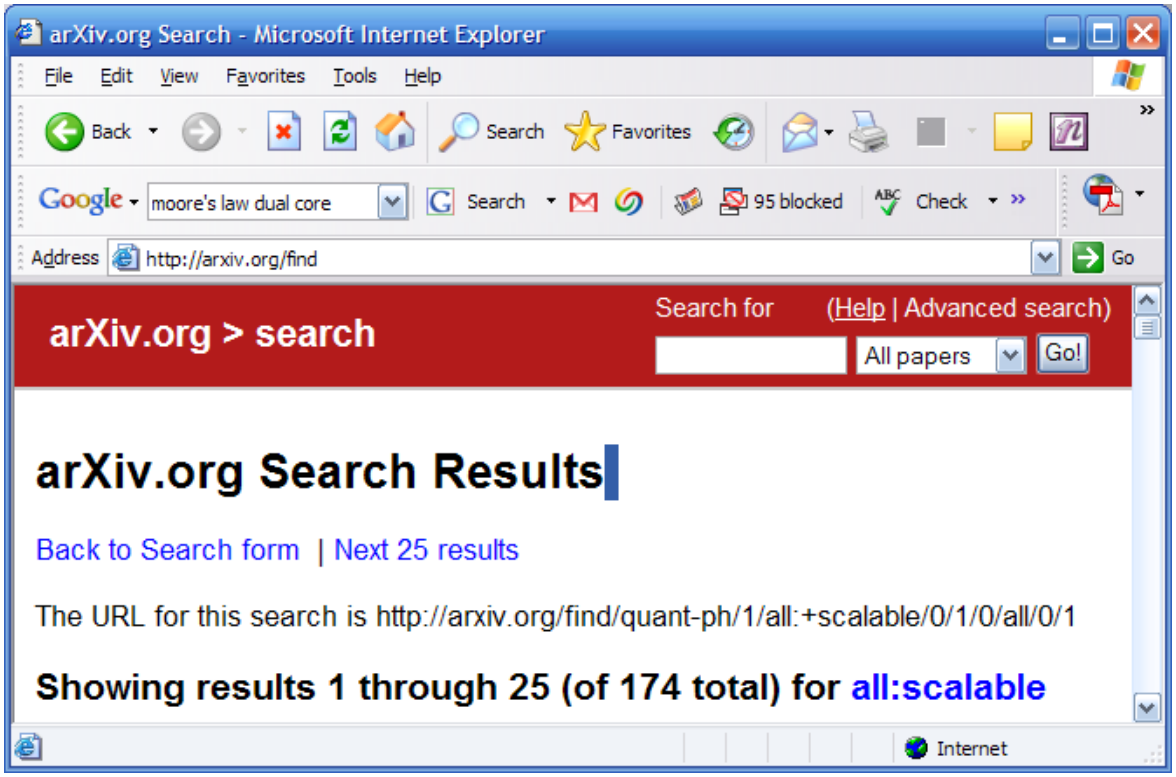
- Toy models which error correct



- Subsystem codes



What Does “Scalable” Mean?



1. All of the necessary components.
2. Easy to fabricate.
3. Below “threshold”

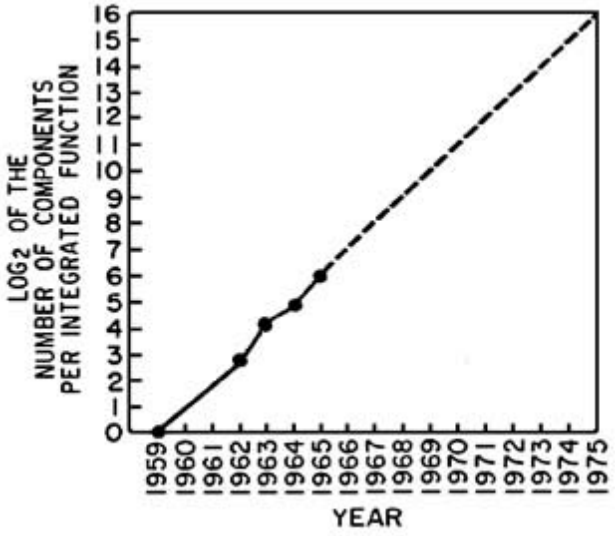
1+3. All of the necessary components which operate below “threshold”

1+2+3. All of the necessary components, below “threshold”, and easy to fabricate.

What Does “Scalable” Mean?



Gordon Moore



Scalable has a rate and a lifetime.

Scalable means that you can predict from current technology this rate and a lower bound on the lifetime.

Scalable means that you can write Moore’s 1965 paper

….and be right!

Topological Codes and Subsystem Codes and Why We Should Care About Them…

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Unive ttle, WA



